1.Indicator

1.1 Definition

For the indicator X of event A,we have:

$$X = \begin{cases} 1 \quad if \quad A \quad occurs \\ 0 \quad otherwise \end{cases}$$

1.2 Expectation

$$\mathbf{E}[X] = 1 \cdot \mathbf{Pr}(A) + 0 \cdot \mathbf{Pr}(A^c) = \mathbf{Pr}(A)$$

1.3 Variance

We have:

$$\mathbf{E}[X^2] = \mathbf{E}[X]$$

Therefore:

$$\mathbf{Var}(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \mathbf{E}[X] - \mathbf{E}[X]^2 = \mathbf{Pr}(A) - \mathbf{Pr}(A)^2 = \mathbf{Pr}(A)(1 - \mathbf{Pr}(A))$$

1.4 Covariance

Let X_A and X_B are indicators of events A and B,we have:

$$\mathbf{Cov}(X_A, X_B) = \mathbf{E}[X_A X_B] - \mathbf{E}[X_A]\mathbf{E}[X_B] = \mathbf{Pr}(AB) - \mathbf{Pr}(A)\mathbf{Pr}(B)$$

1.5 Applications

Boolean Variables:Union Bound

Inclusive and Exclusive inequality

2.Moment Generating Function

$$\mathbf{E}[X^n] = \frac{d^n}{dt^n} MGF_X(t)|_{t=0}$$

$$MGF_X(t) = \mathbf{E}[e^{tx}] = \begin{cases} \sum_x e^{tx} \cdot \mathbf{Pr}(X = x) & x : discrete \\ \int_x e^{tx} \cdot f(x)dx & x : continuous \end{cases}$$

3.Some Useful Inequalities

3.1 Markov inequality
3.2 Chebyshev inequality
3.3 Cantelli inequality
3.4 Paley-Zygmund inequality
3.5 Hölder's inequality
3.6 Hoeffding's inequality
3.7 Kolmogorov's inequality
3.8 Maximum inequalities

reference:some useful inequalities in machine learning and probability and statistics